



ELECT MAINT- OFF SITE Department
SIMHADRI

Note No :1



Ref:SMTTP/ELECT MAINT-/2023-24/LMI INS/282590

Date:

07/07/2023(17:02:20)

Subject:SCHEDULE REVIEW OF LMI/OGN/OPS/ELEC/009-INSULATOR
PROTECTION IN POLLUTED ATMOSPHERE

**Subject: SCHEDULE REVIEW OF LMI/OGN/OPS/ELEC/009-INSULATOR PROTECTION IN
POLLUTED ATMOSPHERE.**

LMI/OGN/OPS/ELEC/009 on Station Preparedness for Lightning & Thunderstorm is reviewed as per
schedule and INSULATOR PROTECTION IN POLLUTED ATMOSPHERE.

No changes are envisaged.

Review proposal of LMI is submitted for the kind approval.

Approving authority for above proposal is HEAD OF THE PROJECT

Submitted by:

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On:07/07/2023(17:02:20) **Ref:**SMTTP/ELECT MAINT-/2023-24/LMI INS/282590

Note No:2

No changes envisage.

Periodic review

Submitted by:

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On:08/07/2023(13:30:33) **Ref:**SMTPP/ELECT MAINT-/2023-24/LMI INS/282590

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On:10/07/2023(09:57:15) **Ref:**SMTPP/ELECT MAINT-/2023-24/LMI INS/282590

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On:10/07/2023(11:45:26) **Ref:**SMTTP/ELECT MAINT-/2023-24/LMI INS/282590

Note No:6

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On:12/07/2023(09:38:16) **Ref:**SMTTP/ELECT MAINT-/2023-24/LMI INS/282590

Note No:7

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Note No:8

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On:15/07/2023(20:24:38) **Ref:**SMTPP/ELECT MAINT-/2023-24/LMI INS/282590

NTPC Limited

SIMHADRI SUPER THERMAL POWER STATION

LMI/OGN/OPS/ELEC/009
ISSUE NO: 1, REVISION: 4
DATE: JULY 2023

**INSULATOR PROTECTION
IN
POLLUTED ATMOSPHERE**

APPROVED FOR

IMPLEMENTATION BY - - - - -

HEAD OF THE PROJECT (SIMHADRI)

Date: _____

NTPC Limited
SIMHADRI SUPER THERMAL POWER STATION
OPERATION & MAINTENANCE DEPARTMENT

LMI (Name) : INSULATOR PROTECTION IN POLLUTED ATMOSPHERE			
LMI No. : LMI /OGN/OPS/ELEC/009			
Issue No.	1	Revision : 4	Date: JULY 2023

	Name	Designation	Signature
Prepared by	J.S.N.V PRASAD	SR. MANAGER(EM)	
Verified by	T.VIJAY KRISHNA	AGM(EM)	
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Reviewed by	D.P.PATRA	GM (O&M)	

INSULATOR PROTECTION IN POLLUTED ATMOSPHERE

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1.0 OBJECTIVE :-

Polluted atmosphere mainly consists of moisture and airborne contaminants like salt, dust etc. Wind is the mechanism by which contaminants are deposited on the insulator surface. Contamination levels can continue to grow over the time unless abated by natural cleaning mechanism like heavy rain. For substation and transmission lines nearer to sea shore, salty water spray causes salt deposition on the insulators. Water condenses on the insulator when the insulator surface cools below the dew point temperature. This behavior is most likely to occur late at night or early in morning. Salty contamination along with moisture provides a path for leakage current to flow thereby causing tracking / flashovers of the insulators. This tracking / flashover causes damage and finally leads to the failure of the insulator.

Flashovers of insulators can cause grid disturbance, power interruptions, damage to generating units and stresses the electrical equipment's.

The objective of this LMI is to protect the insulators in the polluted atmosphere from flashovers there by to assist in maintaining the stability of the grid.

2.0 SCOPE:

This LMI document covers

- 400 KV overhead insulators, post and support insulators.
- Pin and disc insulators in 33 KV Sweet Water, Ash Water and Sea Water lines and post and support insulators in 33KV outdoor substations of sea water, sweet water, and ash water recirculation pump house.
- 25MW Floating solar 33KV metering yard.

3.0 REFERENCE DOCUMENT:

Operation Guidance Note OGN/OPS/ELEC/009

4.0 SUPERSEDED DOCUMENTS:

LMI /OGN/OPS/ELEC/009 Issue: 1 Rev: 2 Dated: April 2010

5.0 **SPECIFIC INSTRUCTIONS TO BE FOLLOWED**

(a) For 400 KV Switchyard

Activity	Responsibility
(a) Visual inspection (b) Manual cleaning (c) Hot line washing	Area Engineer

(b) For 33KV Lines, Solar metering yard and Substation

Activity	Responsibility
(a) Visual inspection (b) Manual cleaning (c) Hotline washing	Area Engineer

6.0 **METHODOLOGY OF IMPLEMENTATION:**

- (a) Visual inspection and Thermography will be carried out once in a month.
- (b) Identify the insulators with tracking /corona discharges during the condition monitoring.
- (c) Clean manually the affected insulators as and when opportunity is available.
- (d) Hot line washing all 400kV and 33kV elements will be done as twice in a year based on the atmospheric conditions as recommended by SRPC

7.0 **RESPONSIBILITY CENTRE FOR IMPLEMENTATION:**

Respective area engineers are responsible for implementation and overall responsibility lies with sectional heads.

- a) Measurement of DM water conductivity and should be $\leq 1 \mu\text{S} / \text{cm}$
- b) Deployed Skilled manpower should be trained from NPTI.
- c) Handheld high-pressure jets must be earthed before start of the hot line washing.
- Responsibility lies with Area engineer.

8.0 **MEASURING EFFECTIVENESS OF HOTLINE WASHING:**

Effective measurement of hotline washing is done by Measurement of turbidity before and after the hotline washing of an equipment wherever possible.

9.0 EXCEPTION REPORTING AND DISPENSATION OF DEVIATIONS

Any exceptions / deviations shall be reported to HOD (EM).

10.0 RECORD KEEPING:

Records will be maintained by concerned area maintenance engineers for switchyard area and 33KV lines in hotline washing register.

11.0 REVIEW:

This LMI document will be reviewed once in every two years by Head of the Project, NTPC Simhadri.